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WEEK 1 LAB – FIRST LLM API CALL USING GOOGLE GEMINI

1. Aim

To connect a Python application to Google's Gemini Large Language Model (LLM) using the Gemini API and generate AI responses through a Python program.

2. Learning Objectives

- Understand the role of an API key in connecting applications to LLMs.
- Set up and use a Python virtual environment.
- Install and use the Google GenAI Python SDK.
- Store API keys securely using a `.env` file.
- Send prompts to Gemini and display generated responses.
- Understand basic API errors and troubleshooting.

3. Software and Tools Required

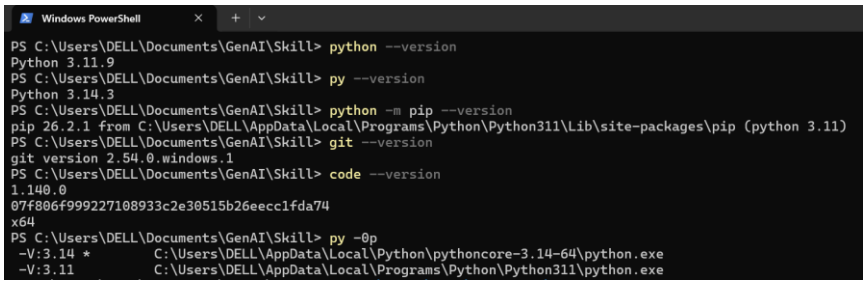
Requirement	Technology
Operating System	Windows
IDE	Visual Studio Code
Programming Language	Python
LLM Provider	Google Gemini
Python SDK	google-genai
Environment Variables	python-dotenv
API Key Platform	Google AI Studio

4. Procedure

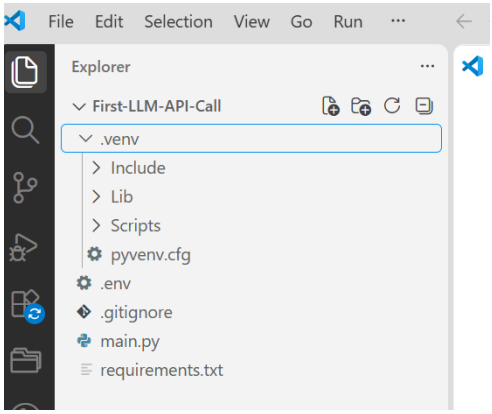
1. Created a separate project folder named `First-LLM-API-Call` inside `Skill/Week - 1`.
2. Checked the Python, pip and Git installations.
3. Created and activated a Python virtual environment (`.venv`).
4. Installed the required packages: `google-genai` and `python-dotenv`.
5. Generated a Gemini API key using Google AI Studio.
6. Created a `.env` file to securely store the API key.
7. Created a `.gitignore` file to prevent sensitive and unnecessary files from being uploaded to GitHub.
8. Developed `main.py` to initialize the Gemini client, send prompts and display AI-generated responses.
9. Executed the Python program and tested different prompts.
10. Verified the generated responses in the VS Code terminal.

5. Implementation and Execution

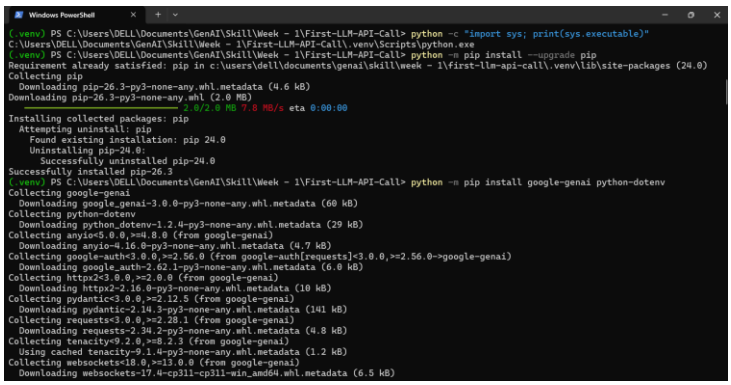
1 – Python Installation Verification



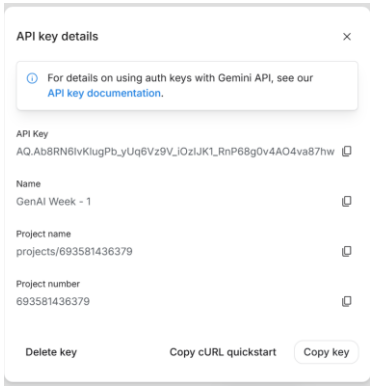
2 – Project Folder Creation



3 – Package Installation



4 – Gemini API Key Setup



6 – main.py code

```python
import os
from dotenv import load_dotenv
from google import genai

# Step 1: Load environment variables from .env
load_dotenv()

# Step 2: Read Gemini API key
api_key = os.getenv("GEMINI_API_KEY")

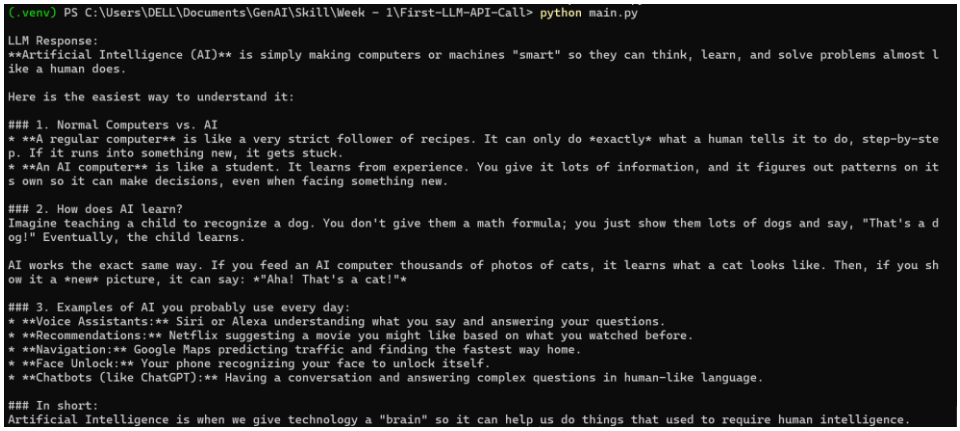
if not api_key:
    raise ValueError(
        "GEMINI_API_KEY not found. Check your .env file."
    )

# Step 3: Initialize Gemini client
client = genai.Client(api_key=api_key)

# Step 4: Send prompt to Gemini model
response = client.models.generate_content(
    model="gemini-3.5-flash",
    contents="What is Artificial Intelligence? Explain in simple words."
)

# Step 5: Display generated response
print("\nLLM Response:")
print(response.text)
```

7 – Successful Gemini API Response



```
(.venv) PS C:\Users\DELL\Documents\GenAI\Skill\Week - 1\First-LLM-API-Call> python main.py

LLM Response:
**Artificial Intelligence (AI)** is simply making computers or machines "smart" so they can think, learn, and solve problems almost l
ike a human does.

Here is the easiest way to understand it:

### 1. Normal Computers vs. AI
* **A regular computer** is like a very strict follower of recipes. It can only do *exactly* what a human tells it to do, step-by-ste
p. If it runs into something new, it gets stuck.
* **An AI computer** is like a student. It learns from experience. You give it lots of information, and it figures out patterns on it
s own so it can make decisions, even when facing something new.

### 2. How does AI learn?
Imagine teaching a child to recognize a dog. You don't give them a math formula; you just show them lots of dogs and say, "That's a d
og!" Eventually, the child learns.

AI works the exact same way. If you feed an AI computer thousands of photos of cats, it learns what a cat looks like. Then, if you sh
ow it a *new* picture, it can say: *"Aha! That's a cat!"*

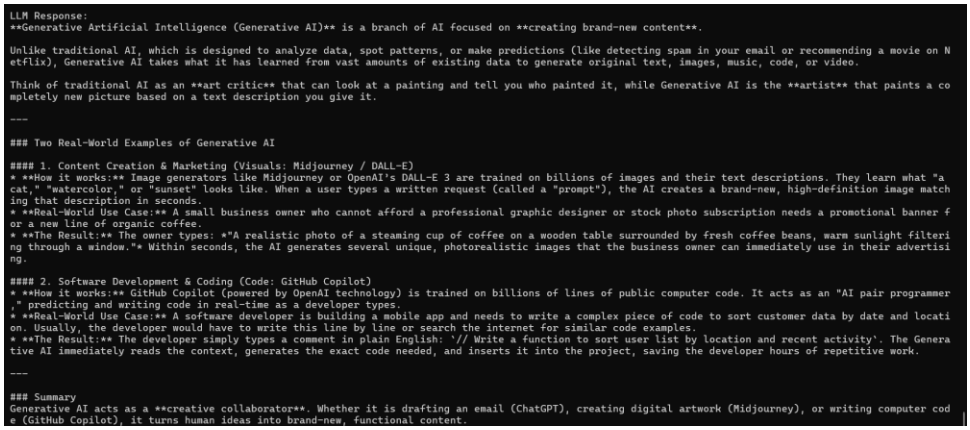
### 3. Examples of AI you probably use every day:
* **Voice Assistants:** Siri or Alexa understanding what you say and answering your questions.
* **Recommendations:** Netflix suggesting a movie you might like based on what you watched before.
* **Navigation:** Google Maps predicting traffic and finding the fastest way home.
* **Face Unlock:** Your phone recognizing your face to unlock itself.
* **Chatbots (like ChatGPT):** Having a conversation and answering complex questions in human-like language.

### In short:
Artificial Intelligence is when we give technology a "brain" so it can help us do things that used to require human intelligence.
```

8 – Different Prompt Outputs

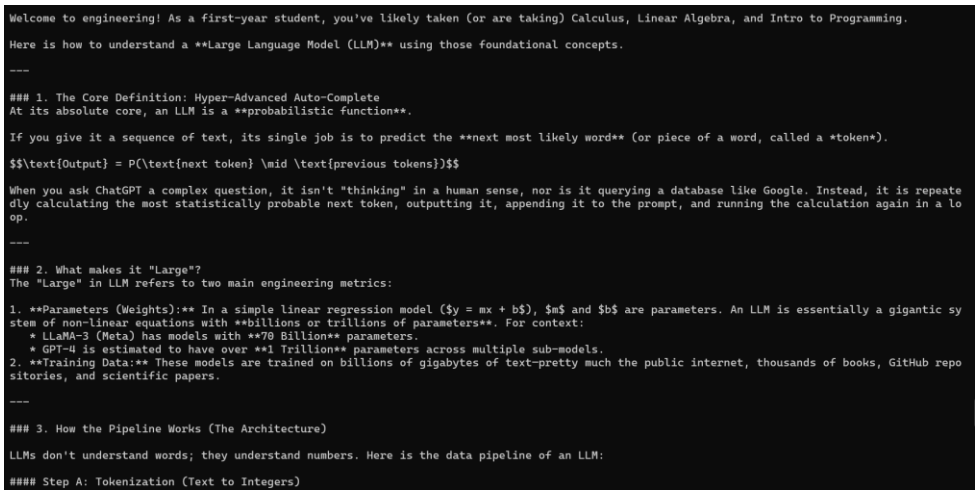
Prompt – 1

contents="Explain Generative AI with two real-world examples."



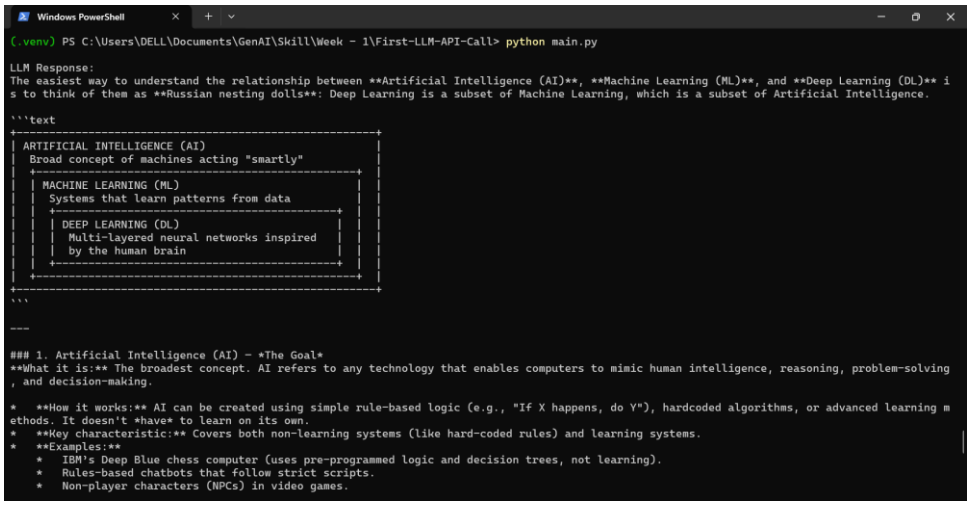
Prompt – 2

contents="What is a Large Language Model? Explain for a first-year engineering student."



Prompt – 3

contents="Differentiate Artificial Intelligence, Machine Learning, and Deep Learning."



6. Result

Successfully created and executed a Python application connected to Google's Gemini API. The application securely accessed the API key through the `.env` file, sent prompts to the Gemini model and displayed AI-generated responses in the terminal.

7. Conclusion

This experiment provided practical knowledge of connecting Python applications with Large Language Models using APIs. It also introduced virtual environments, package management, API key security and prompt experimentation.

8. Key Learnings

- An API enables communication between two software applications.
- An API key is used to authenticate requests.
- A virtual environment keeps project dependencies separate.
- The `.env` file helps protect sensitive credentials.
- Prompts determine the instructions given to an LLM.
- LLM responses are generated dynamically and may vary.